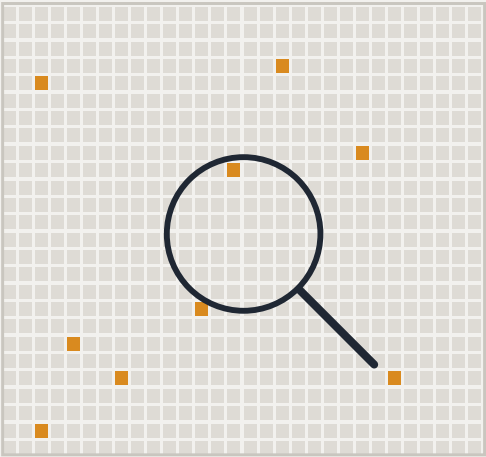


From 6K unlabeled SAE atoms to the five a probe really uses — one atom switched at a time, ranked by verified counterfactual flips

(a) Today: browse an SAE

manual · correlational · not actionable



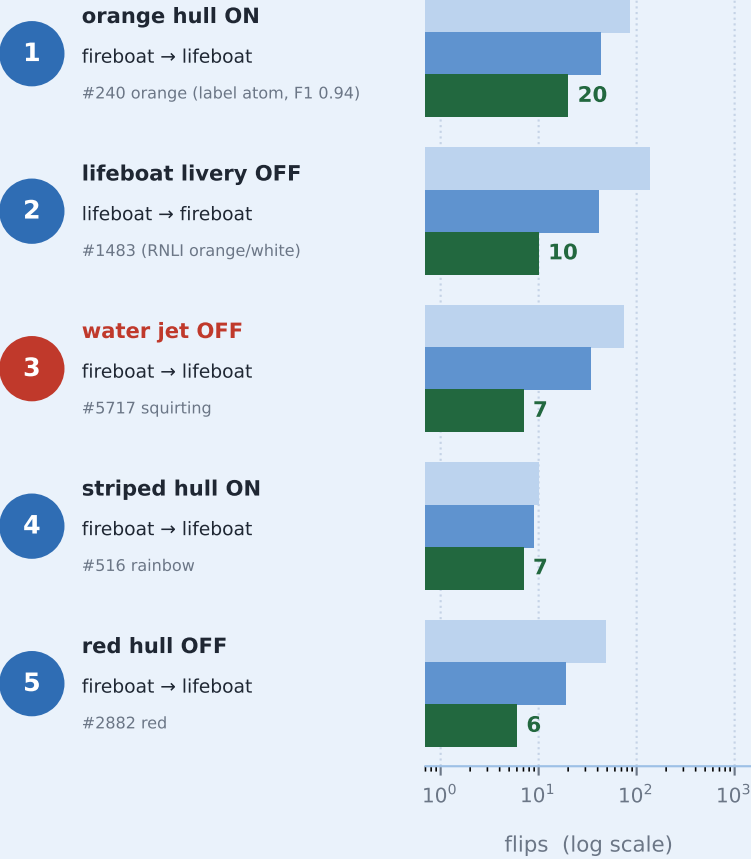
per atom: highest-activating images



✗ jet or boat? unclear · no fix

(b) Rank all directions by verified flips

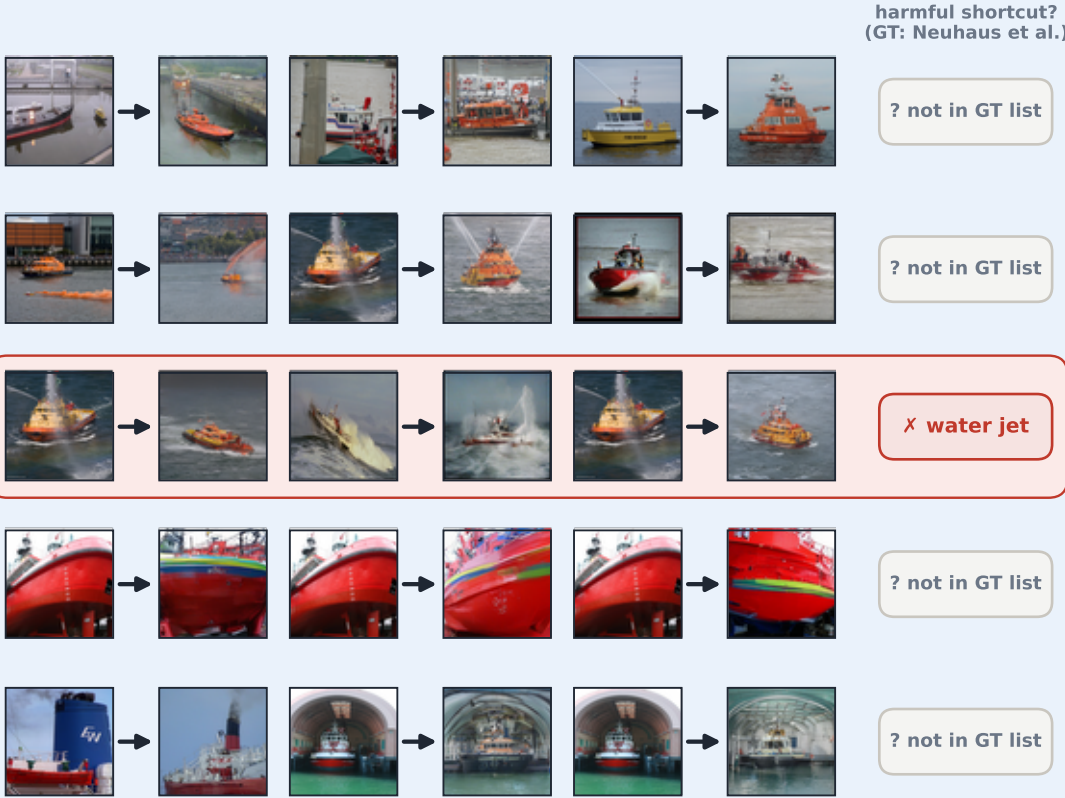
DINOv3 ViT-L/16 linear probe, ImageNet fireboat vs lifeboat (99.2 % test acc.)  
public MSAE on CLIP ViT-L/14 (6,144 atoms), no SAE training  
2,340 images swept · one atom driven to its bound · 46,903 renders (fresh noise)



light blue: latent flips (distilled probe)  
dark blue: ambient flips (real probe on decoded image)  
dark green: verified flips — rank key

(c) Counterfactuals of the top-5 only (3 verified flips per direction)

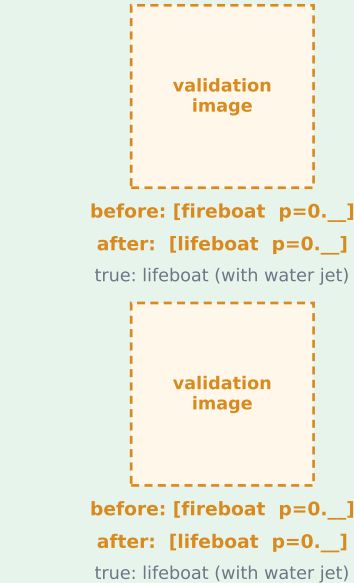
original  $\mathbf{x}$  → counterfactual  $\tilde{\mathbf{x}} = \text{decode}(\mathbf{z} + \delta \mathbf{v}_k)$  — the probe's decision flips on every pair



single-atom sweep (no concept replacement, fresh-noise renderer): the water jet of Neuhaus et al. (2022) is rank 3 with 7 verified flips — switching #5717 off removes the spray and the probe says lifeboat. The hull-colour atoms around it are not in the paper's list; whether they are shortcuts is the teacher's call (not judged in this run). Fresh noise regenerates the scene, so pairs differ in layout; the guided single-atom renderer flipped too few images to rank.

(d) Fix the probe

CFKD on the  $\mathbf{x}$  directions:  
validation images before → after



CFKD run pending  
(reproduce\_didae\_results.sh, last block)

✓ backbone untouched